File and Directory Permission In Linux

Akintunde Ridwan Oladimeji
Student ID : 2025/INT/11578

Course Name : INT305 Network Security and Protocols

Instructor Name : Mr Aminu Idris (AMCPN)

Date of Submission : 04/02/2026
Version : 1.0

Table of Contents

1.0 Introduction

2.0 Aim and objectives

2.1 Tools

3.0 Methodology

3.1 Create Directory

3.2 Managing Directory Permissions

3.3 Changing Directory Ownership

3.4 Check Directory Permissions

4.0 Results and Analysis

5.0 Conclusion and Recommendation

Executive Summary

This lab focused on exploring file and directory permission in the Linux operating system. Key activities include creating a directory and modifying the directory permission and ownership. The lab revealed that a new directory created in Linux has a default permission usually **755 (drwxr-xr-x)** or **775 (drwxrwxr-x)** until an administrator modifies the file or directory permission.

1.0 Introduction

File and directory permissions in Linux are a key security feature used to control how users interact with system resources. Every file and directory is assigned an owner, a group, and permission settings that determine who can read, write, or execute the resource. Permissions are divided into three categories: user (owner), group, and others, with each category having read (view content), write (modify content), and execute (run file or access directory) privileges.

2.0 Aim and objectives

The aim of this lab is to understand how to set and manage permissions for files and directories in Linux.

The Objectives of this lab is to

- create a directory
- modify and confirm the permission
- modify the directory ownership

2.1 Tools

- Kali Linux

3.0 Methodology

3.1 Create Directory

Command : **`mkdir my_project`**

Purpose: Used to create new working directories.

3.2 Managing Directory Permissions

Set Directory Permission

Command : **`chmod 755 directoryname`**

Command : **`chmod 755 my_project`**

Purpose : Allows full control for owner, Read and execute access for group and others

3.3 Changing Directory Ownership

Command **: `chown -R owner:group directoryname`**

Command **: `chown student1:students my_project`**

Purpose : Changes Directory owner , Directory group, Applies changes recursively to subdirectories and files.

3.4 Check Directory Permissions

Command **: `ls -ld directoryname`**

Command **: `ls -ld my_project`**

Purpose : Displays permissions of the directory itself rather than its contents

.

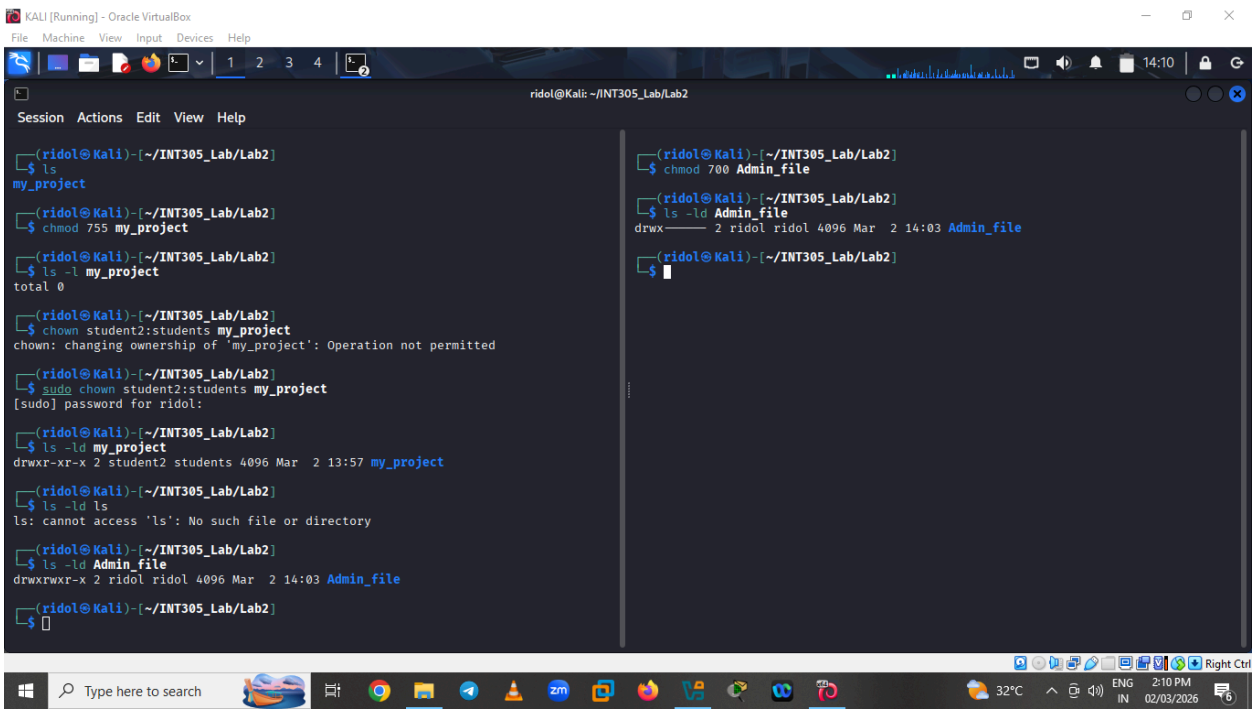

Fig 1: Evidence showing directory ownership and permission settings on a Linux terminal

4.0 Results and Analysis

When a new directory is created in Linux, the default permission is usually **755 (drwxr-xr-x)** or **775 (drwxrwxr-x)** until an administrator modify the file or directory permission just as observed in the directory **Admin_file** from the lab.

5.0 Conclusion and Recommendation

The lab emphasises on the importance of file and directory permission management for access control. This is in accordance with the principles of least privilege alongside ISO compliance practices.